

DATA MINING & DATA WAREHOUSING

EXPERIMENTS 1 TO 7 — COMPLETE THEORY

TCET — Department of Computer Engineering

This document contains the theory required for Experiments 1–7, organized for practical-record preparation and viva revision. Code and detailed numerical examples are intentionally excluded.

Experiments Covered

Experiment	Topic
1	Data Warehouse / Data Mart and Dimensional Data Modeling
2	OLAP Operations — Slice, Dice, Drill-Down, Roll-Up and Pivot
3	Classification using Decision Tree (ID3/C4.5 concepts)
4	Linear Regression
5	Clustering using K-Means
6	Hierarchical Clustering
7	Apriori Algorithm for Market Basket Analysis

EXPERIMENT 1 — DATA WAREHOUSE & DIMENSIONAL MODELING

1. Data Warehouse

A Data Warehouse (DW) is a centralized repository designed to store integrated data collected from multiple operational and external sources. Unlike an operational database, which is optimized for day-to-day transactions, a data warehouse is optimized for reporting, analytical processing and decision-making.

A data warehouse generally contains historical, subject-oriented and integrated information. Data from different source systems is extracted, transformed and loaded into the warehouse through an ETL process. The stored data can then be queried by analysts and business-intelligence tools.

2. Characteristics of a Data Warehouse

- Subject-oriented: data is organized around important business subjects such as sales, customers or products.
- Integrated: data from different systems is converted into a consistent format.
- Time-variant: historical data is retained so that changes can be analyzed over time.
- Non-volatile: warehouse data is primarily read and analyzed rather than continuously updated like an operational system.

3. Data Mart

A Data Mart is a smaller, subject-specific subset of a data warehouse. It is designed for a particular department or business area such as sales, finance, marketing or human resources. A data mart provides faster and more focused access to data for a specific group of users.

4. Data Warehouse Architecture

A typical architecture contains a source layer, ETL layer, storage layer and presentation layer. Operational databases and files act as sources. ETL extracts, cleans and transforms the data before loading it into the warehouse. The presentation layer exposes the data through reports, dashboards and OLAP tools.

5. Dimensional Data Modeling

Dimensional modeling is a design technique used for analytical databases. It organizes information around measurable business events called facts and descriptive entities called dimensions. The central fact table contains measures, while dimension tables provide context for analyzing those measures.

6. Fact Table

A fact table stores quantitative measurements of a business process. Examples include quantity sold, sales amount, profit and discount. It normally contains foreign keys connecting it to dimension tables.

7. Dimension Table

A dimension table stores descriptive attributes used to filter, group and analyze facts. Common dimensions include Date, Product, Customer and Store.

8. Star Schema

In a star schema, one central fact table is directly connected to multiple denormalized dimension tables. The structure resembles a star. It is simple, easy to query and commonly used for OLAP workloads.

9. Snowflake Schema

A snowflake schema is an extension of the star schema in which dimension tables are normalized into additional related tables. It reduces redundancy but increases the number of joins and makes the structure more complex.

10. Fact Constellation Schema

A fact constellation, also called a galaxy schema, contains multiple fact tables that share dimension tables. It is useful when an organization has several related business processes, such as Sales and Returns, that use common Product, Customer and Date dimensions.

11. ETL Process

ETL stands for Extract, Transform and Load. Extract collects data from source systems. Transform cleans, validates, converts and integrates the data. Load places the transformed data into the warehouse.

12. OLAP

Online Analytical Processing (OLAP) provides multidimensional analysis of warehouse data. It enables users to examine measures across dimensions such as time, product, location and customer.

EXPERIMENT 2 — OLAP OPERATIONS

OLAP operations allow users to view and analyze multidimensional warehouse data from different levels and perspectives.

1. Slice

Slice selects a single value from one dimension of a multidimensional data cube, producing a smaller sub-cube. For example, selecting Year = 2025 from a sales cube removes the other years and focuses the analysis only on 2025.

2. Dice

Dice selects multiple values or ranges from two or more dimensions. It produces a smaller sub-cube based on several conditions. For example, selecting Products = Laptop and Mouse, Cities = Mumbai and Pune, and Year = 2025 is a dice operation.

3. Drill-Down

Drill-down moves from summarized, higher-level data to more detailed data. For example, a report may move from Year → Quarter → Month → Day. It helps users investigate the reasons behind an aggregate value.

4. Roll-Up

Roll-up is the opposite of drill-down. It moves from detailed data to a higher level of aggregation. For example, Day → Month → Quarter → Year. It reduces detail and provides a summarized view.

5. Pivot

Pivot, or rotate, changes the orientation of dimensions in a report or cube. It does not necessarily change the underlying data; instead, it changes how the dimensions are displayed so that another relationship becomes easier to inspect.

6. OLAP Cube

An OLAP cube is a multidimensional representation of data. Dimensions describe the context of analysis, while measures are numerical values such as sales amount, quantity or profit.

7. Difference Between Operations

Operation	Main purpose
Slice	Select one value of a dimension.
Dice	Select multiple values/ranges across dimensions.
Drill-Down	Move to greater detail.
Roll-Up	Move to a higher summary level.
Pivot	Rotate/reorient the analytical view.

8. Importance of OLAP

- Supports fast multidimensional analysis.
- Helps identify trends and patterns.
- Allows comparison across time, geography and products.
- Supports business reporting and decision-making.
- Allows users to move between summary and detailed views.

EXPERIMENT 3 — CLASSIFICATION USING DECISION TREE

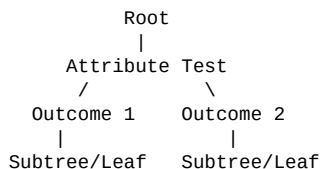
1. Classification

Classification is a supervised machine-learning technique used to assign an input record to one of several predefined classes. A model is trained using labeled examples and then used to predict the class of new records.

2. Decision Tree

A decision tree is a tree-like model in which internal nodes represent tests on attributes, branches represent outcomes of those tests, and leaf nodes represent class labels. Classification follows a path from the root to a leaf according to the attribute values of the input record.

3. Decision Tree Structure



4. Root Node and Leaf Node

The root is the first attribute selected for splitting the dataset. Internal nodes perform further tests. A leaf node contains the final predicted class.

5. Entropy

Entropy measures the impurity or uncertainty of a dataset. A dataset containing examples from only one class has zero entropy. Higher entropy indicates a more mixed class distribution.

$$\text{Entropy}(S) = - \sum p(i) \log_2(p(i))$$

6. Information Gain

Information Gain measures how much an attribute reduces uncertainty after splitting the dataset. ID3 selects the attribute with the highest information gain.

$$\text{Gain}(S,A) = \text{Entropy}(S) - \sum [|S_v| / |S| \times \text{Entropy}(S_v)]$$

7. ID3

ID3 is a decision-tree induction algorithm that uses entropy and information gain to choose splitting attributes. It recursively partitions the data until suitable stopping conditions are reached.

8. C4.5

C4.5 extends ID3 with improvements such as gain ratio, handling of continuous attributes, pruning and better treatment of missing values.

9. Classification Process

- Collect labeled training data.
- Calculate impurity of the current dataset.
- Evaluate candidate attributes.
- Select the best splitting attribute.
- Partition the dataset.
- Repeat recursively for the resulting subsets.
- Stop when a suitable leaf/class is obtained.

10. Advantages and Limitations

Decision trees are easy to understand, visualize and convert into rules. They can handle categorical attributes and are useful for exploratory analysis. However, an unpruned tree may overfit, and small changes in training data can sometimes produce a different tree.

EXPERIMENT 4 — LINEAR REGRESSION

1. Linear Regression

Linear regression is a supervised statistical and machine-learning technique used to model the relationship between a numerical dependent variable and one or more independent variables. Simple linear regression uses one independent variable and fits a straight line through the observations.

$$Y = \alpha + \beta X$$

Y = predicted dependent variable

X = independent variable

α = intercept

β = slope

2. Slope

The slope represents the expected change in Y for a one-unit increase in X. A positive slope indicates an increasing relationship, while a negative slope indicates a decreasing relationship.

$$\beta = \frac{\sum[(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum[(X_i - \bar{X})^2]}$$

$$\alpha = \bar{Y} - \beta \bar{X}$$

3. Least Squares

The ordinary least squares method chooses the line that minimizes the sum of squared differences between observed and predicted values.

$$\text{Minimize } \sum(Y_i - \hat{Y}_i)^2$$

4. Residual

A residual is the difference between an observed value and the value predicted by the regression model.

$$\text{Residual} = \text{Actual } Y - \text{Predicted } Y$$

5. Prediction

After α and β are calculated, a new input value can be substituted into the regression equation to estimate the corresponding output.

6. R²

R², or coefficient of determination, describes the proportion of variation in the dependent variable explained by the regression model.

$$R^2 = 1 - \text{SSE}/\text{SST}$$

7. Correlation vs Regression

Correlation	Regression
Measures association.	Builds a predictive relationship.
No explicit dependent/independent role.	Defines predictor and response.
Summarized by r.	Produces a regression equation.

8. Applications

- Salary prediction from experience.
- Sales prediction from advertising.
- House-price estimation.
- Demand forecasting.
- Prediction of marks or performance.

9. Limitations

Linear regression assumes an approximately linear relationship. Outliers can strongly affect the fitted line, and extrapolation beyond the observed range can be unreliable. A high R² alone does not establish causation.

EXPERIMENT 5 — K-MEANS CLUSTERING

1. Clustering

Clustering is an unsupervised learning technique that groups observations according to similarity without requiring predefined class labels. It is used in customer segmentation, pattern recognition, image analysis and exploratory data mining.

2. K-Means

K-Means is a partitioning clustering algorithm that divides observations into K clusters. Each cluster is represented by a centroid, which is the mean of the points assigned to that cluster.

3. K-Means Algorithm

1. Choose K.
2. Select initial centroids.
3. Calculate distance from each point to every centroid.
4. Assign each point to its nearest centroid.
5. Recalculate each centroid as the mean of its cluster.
6. Repeat until assignments/centroids stabilize.

4. Euclidean Distance

$$d(P, Q) = \sqrt{\sum (P_i - Q_i)^2}$$

5. Centroid

A centroid is the mean position of all observations belonging to a cluster. For numerical features, each coordinate of the centroid is obtained by averaging the corresponding feature values.

Centroid = Mean of all points in the cluster

6. Objective Function

K-Means attempts to minimize the within-cluster sum of squared distances, often called the within-cluster sum of squares (WCSS).

$$WCSS = \sum \text{over clusters} \sum \text{over points} ||x - \text{centroid}||^2$$

7. Choosing K

The number of clusters K must generally be selected before the basic K-Means algorithm starts. The elbow method and silhouette analysis are commonly used to help choose a suitable value.

8. Advantages

- Simple and relatively fast.
- Easy to implement and interpret.
- Works well for compact, well-separated numerical clusters.

9. Limitations

- The value of K must be chosen.
- Sensitive to initial centroids.
- Sensitive to outliers and feature scaling.
- May perform poorly for non-spherical or unevenly sized clusters.

EXPERIMENT 6 — HIERARCHICAL CLUSTERING

1. Hierarchical Clustering

Hierarchical clustering builds a hierarchy of clusters rather than directly producing only one partition. The hierarchy is commonly visualized using a dendrogram, which shows the sequence and distance of cluster merges or splits.

2. Agglomerative and Divisive Methods

Method	Description
Agglomerative	Bottom-up: each observation starts separately and clusters are repeatedly merged.
Divisive	Top-down: all observations start together and clusters are recursively divided.

3. Agglomerative Algorithm

1. Start with each point as an individual cluster.
2. Calculate distances between clusters.
3. Find the closest pair.
4. Merge the closest clusters.
5. Update cluster distances.
6. Repeat until the desired hierarchy/number of clusters is obtained.

4. Linkage

Linkage defines the distance between two clusters.

Linkage	Definition
Single	Minimum pairwise distance.
Complete	Maximum pairwise distance.
Average	Average pairwise distance.
Ward	Merge causing the smallest increase in within-cluster variance.

5. Single Linkage

The experiment uses single linkage. The distance between two clusters is the minimum distance between any point in the first cluster and any point in the second cluster.

$$d(A,B) = \min d(p,q), \quad p \in A \text{ and } q \in B$$

6. Dendrogram

A dendrogram is a tree diagram representing the hierarchy. The height of a merge represents the distance at which the clusters were joined. Cutting the dendrogram at a selected height produces a desired set of clusters.

7. Advantages

- Shows a complete hierarchy of possible groupings.
- Does not require a fixed K before constructing the hierarchy.
- Dendrogram provides an intuitive visual representation.

8. Limitations

- Can be computationally expensive for large datasets.
- Standard agglomerative merging cannot normally be undone.
- Sensitive to distance metric, linkage method and feature scaling.

9. Applications

- Customer segmentation.
- Document and image grouping.
- Bioinformatics.
- Pattern recognition.
- Exploratory analysis.

EXPERIMENT 7 — APRIORI ALGORITHM FOR MARKET BASKET ANALYSIS

1. Apriori Algorithm

Apriori is a classic association-rule mining algorithm used to discover frequent itemsets and association patterns in transactional databases. It works level by level: frequent 1-itemsets are used to generate candidates for 2-itemsets, then 3-itemsets, and so on.

2. Market Basket Analysis

Market Basket Analysis identifies products or events that frequently occur together. For example, a retailer may discover that customers who purchase one product often purchase another. These relationships can support recommendations, cross-selling and product placement.

3. Important Terms

Term	Meaning
Item	A single product or element.
Itemset	A set of one or more items.
Transaction	A collection of items occurring together.
Frequent itemset	An itemset satisfying the minimum support threshold.
Support	Frequency of an itemset in the database.
Confidence	Strength of an association rule.
Lift	Compares observed co-occurrence with expected co-occurrence under independence.

4. Apriori Property

The central Apriori property is: **every non-empty subset of a frequent itemset must also be frequent**. Conversely, if an itemset is not frequent, none of its supersets can be frequent.

If {A,B,C} is frequent:
{A,B}, {A,C}, {B,C}, {A}, {B}, {C}
must also be frequent.

If {A,B} is not frequent:
{A,B,C} cannot be frequent.

5. Support

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

Minimum support is used to eliminate itemsets that occur too rarely.

6. Join and Prune

Apriori uses two important operations. The join step combines frequent (k-1)-itemsets to generate candidate k-itemsets. The prune step removes a candidate if any of its (k-1)-itemset subsets is not frequent.

$$L(k-1) \rightarrow \text{JOIN} \rightarrow C(k) \rightarrow \text{PRUNE} \rightarrow L(k)$$

7. Association Rules

A rule has the form $X \rightarrow Y$, where X and Y are disjoint itemsets. Confidence indicates how frequently Y occurs among transactions containing X.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{[\text{Support}(X) \times \text{Support}(Y)]}$$

8. Apriori Process

Transaction Database
↓
Candidate C1
↓
Support Counting
↓
Frequent L1
↓

```
Join + Prune
  ↓
Candidate C2
  ↓
Support Counting
  ↓
Frequent L2
  ↓
Continue until no frequent itemsets remain
```

9. Advantages

- Simple and easy to understand.
- Uses pruning to reduce the search space.
- Produces interpretable frequent itemsets and rules.
- Useful for retail and recommendation analysis.

10. Limitations

- May require repeated database scans.
- Can generate many candidate itemsets.
- Performance can become poor with low minimum support.
- FP-Growth and other approaches can avoid some candidate-generation overhead.

11. Applications

- Retail market basket analysis.
- Recommendation systems.
- Cross-selling.
- Website click-pattern analysis.
- Fraud and event-pattern discovery.

QUICK VIVA REVISION — EXPERIMENTS 1 TO 7

Exp.	Topic	One-line key point
1	Data Warehouse	Centralized historical repository for analytical processing.
2	OLAP	Slice, dice, drill-down, roll-up and pivot provide multidimensional analysis.
3	Decision Tree	Classification by recursively splitting data using attribute tests.
4	Linear Regression	Fits a line $\hat{y} = \alpha + \beta X$ for numerical prediction.
5	K-Means	Assigns points to nearest centroid and repeatedly updates centroids.
6	Hierarchical	Builds a hierarchy by repeatedly merging or splitting clusters.
7	Apriori	Finds frequent itemsets using support and the downward-closure property.

Essential Formulas

Decision Tree:

$$\text{Entropy}(S) = -\sum p(i) \log_2(p(i))$$

Linear Regression:

$$\hat{y} = \alpha + \beta X$$

$$\beta = \frac{\sum [(X_i - \bar{X})(Y_i - \bar{Y})]}{\sum [(X_i - \bar{X})^2]}$$

$$\alpha = \bar{Y} - \beta \bar{X}$$

K-Means:

$$d(P, Q) = \sqrt{\sum (P_i - Q_i)^2}$$

Hierarchical Single Linkage:

$$d(A, B) = \min d(p, q)$$

Apriori:

$$\text{Support}(X) = \text{count}(X) / \text{total transactions}$$

$$\text{Confidence}(X \rightarrow Y) = \text{Support}(X \cup Y) / \text{Support}(X)$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{[\text{Support}(X) \times \text{Support}(Y)]}$$

End of Theory Notes